GRADE 12 CHEMISTRY

UNIT 2 — ELECTROCHEMISTRY

Redox reactions, electrolysis, Faraday’s laws, industrial electrolysis and voltaic cells

Exam-focused revision booklet

Prepared from the uploaded Ethiopian Grade 12 Chemistry Student Textbook (Ministry of Education, 2023). The explanations below are newly written study notes, not copied textbook pages. Diagrams are original schematic summaries.

How to use this booklet

Read the core idea, memorize the key relationship or equation, redraw the visual summary from memory, then answer both alternative questions without looking. For calculations, show units and working. For process questions, give the stages in the correct order.

CORE CONCEPTS	KEY EQUATIONS / PROCESSES	EXAM PRACTICE

Coverage: 19 focused topic sheets plus this guide page = 20 pages minimum. The two questions on each topic are alternatives for self-testing.

2.01 Meaning of redox reactions

Core concept

Oxidation and reduction occur together in a redox reaction. Oxidation is loss of electrons and an increase in oxidation number; reduction is gain of electrons and a decrease in oxidation number. A reaction can be checked by tracking electron transfer or oxidation-number changes.

Remember

OIL RIG: Oxidation Is Loss; Reduction Is Gain

Meaning of redox reactions	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define oxidation and reduction using electron transfer.

B. Identify whether Mg → Mg²■ + 2e■ is oxidation or reduction.

2.01 • Exam diagram & MCQ practice

Key exam idea: Oxidation is loss of electrons; reduction is gain of electrons. They occur together.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The change $Mg \rightarrow Mg^{2+} + 2e^-$ is:

A. oxidation
B. reduction
C. neutralization
D. precipitation

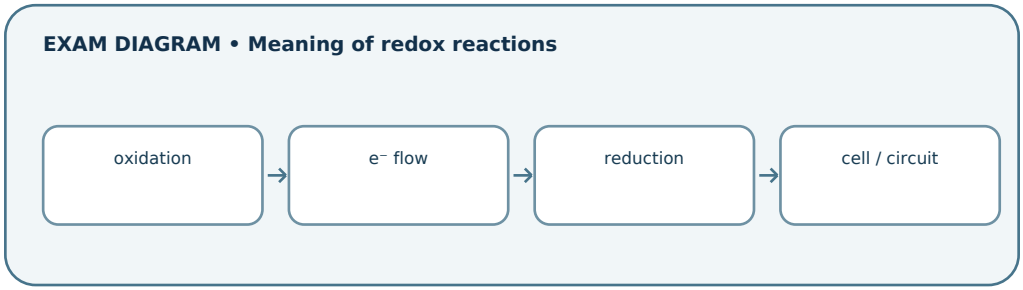
Answer: A. Magnesium loses two electrons, so it is oxidized.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.02 Oxidation numbers

Core concept

Oxidation number is a bookkeeping value used to track electron transfer. An element in its free state has oxidation number zero; a monatomic ion has an oxidation number equal to its charge. The sum of oxidation numbers equals the total charge of the species. Apply standard rules consistently to identify what changes.

Remember

Sum of oxidation numbers = overall charge

Oxidation numbers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Determine oxidation numbers of sulfur in SO■ and SO■²■.

B. Find the oxidation number of Mn in MnO■■.

2.02 • Exam diagram & MCQ practice

Key exam idea: The sum of oxidation numbers equals the overall charge; oxygen is usually −2 and hydrogen usually +1.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. What is the oxidation number of sulfur in SO_4^{2-}?

A. +6
B. +4
C. −2
D. +2

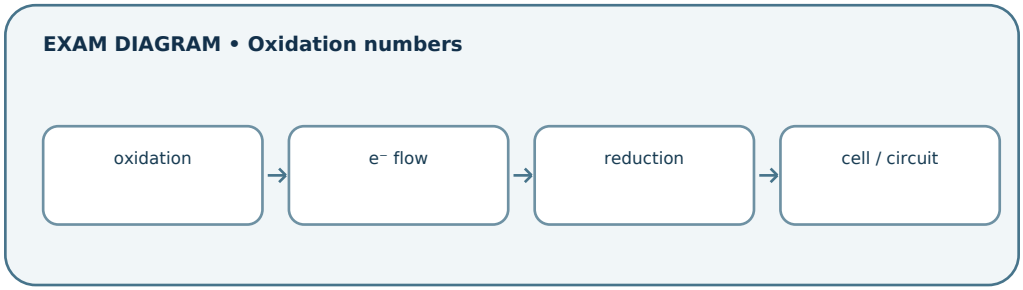
Answer: A. Let sulfur be x: x + 4(−2) = −2, so x = +6.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.03 Oxidizing and reducing agents

Core concept

The oxidizing agent causes another species to be oxidized and is itself reduced. The reducing agent causes another species to be reduced and is itself oxidized. The agent's own oxidation-number change is the reverse of the process it causes. Identify the species gaining electrons first, then name the agents.

Remember

Oxidizing agent is reduced; reducing agent is oxidized

Oxidizing and reducing agents	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. In Zn + Cu^{2}■ → Zn^{2}■ + Cu, identify both agents.

B. Explain why chlorine can act as an oxidizing agent in a redox reaction.

2.03 • Exam diagram & MCQ practice

Key exam idea: An oxidizing agent accepts electrons and is reduced; a reducing agent donates electrons and is oxidized.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, the oxidizing agent is:

A. Cu^{2+}
B. Zn
C. Zn^{2+}
D. Cu

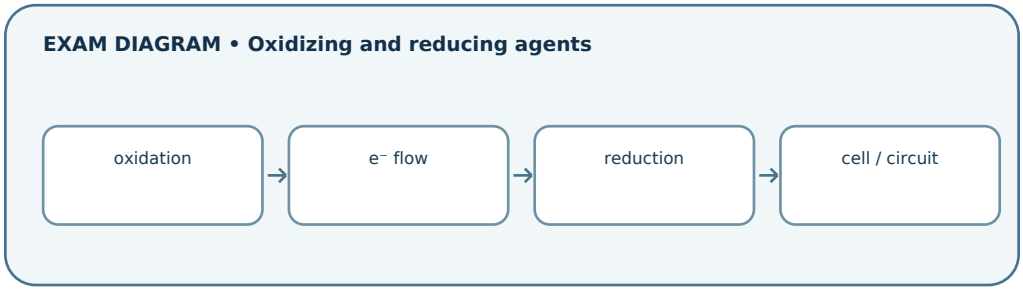
Answer: A. Cu^{2+} gains electrons to form Cu, so it is reduced and acts as the oxidizing agent.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.04 Recognizing redox reactions

Core concept

Compare oxidation numbers before and after reaction. If at least one element's oxidation number increases and another decreases, the reaction is redox. Reactions with no oxidation-number change are not classified as redox by this test. Balance formulas and charges before assigning oxidation states.

Remember

Increase = oxidation; decrease = reduction

Recognizing redox reactions	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Determine whether 2Mg + O■ → 2MgO is redox.

B. Check whether HCl + NaOH → NaCl + H■O is a redox reaction.

2.04 • Exam diagram & MCQ practice

Key exam idea: A reaction is redox when oxidation numbers change; electron transfer is not present in ordinary acid–base neutralization.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which reaction is a redox reaction?

A. $2Mg + O_2 \rightarrow 2MgO$
B. $HCl + NaOH \rightarrow NaCl + H_2O$
C. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
D. $H^+ + OH^- \rightarrow H_2O$

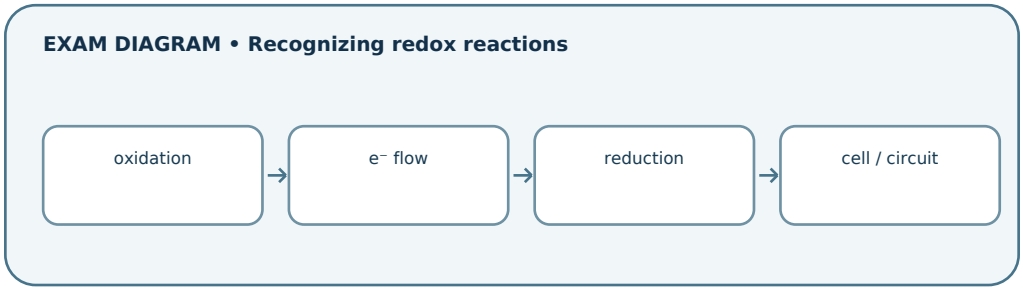
Answer: A. Mg changes from 0 to +2 and oxygen from 0 to −2, showing oxidation and reduction.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.05 Balancing redox equations

Core concept

Redox equations must conserve both atoms and charge. The textbook introduces systematic balancing using oxidation and reduction changes. Separate the oxidation and reduction changes, equalize electron transfer, combine the half-reactions, and check atoms and total charge. In acidic or basic media, balance oxygen and hydrogen with appropriate species as required by the method.

Remember

Electrons lost = electrons gained

Balancing redox equations	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Balance a simple metal-displacement redox equation and verify charge.

B. Explain why electron balance is essential in a redox equation.

2.05 • Exam diagram & MCQ practice

Key exam idea: A balanced redox equation conserves atoms and total charge; electrons lost must equal electrons gained.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why must electrons lost equal electrons gained when balancing a redox equation?

A. To conserve charge
B. To conserve volume
C. To make all coefficients equal
D. To remove the need to balance atoms

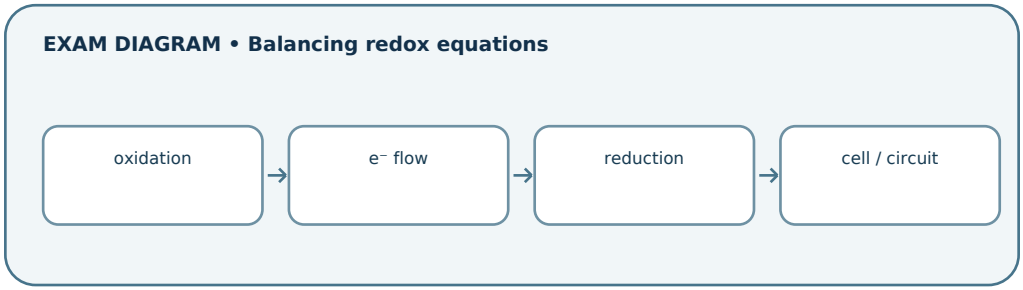
Answer: A. Electron transfer cannot create or destroy net charge; the half-reactions must exchange equal electron numbers.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.06 Electrolytes and electrolysis

Core concept

An electrolyte conducts electricity through mobile ions when molten or in solution. Electrolysis uses electrical energy to drive a non-spontaneous chemical change. Positive ions move toward the cathode and negative ions toward the anode. Reduction occurs at the cathode; oxidation occurs at the anode.

Remember

Cathode: reduction; anode: oxidation

Electrolytes and electrolysis	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define electrolysis and explain the role of ions.

B. State the direction of cation and anion movement in an electrolytic cell.

2.06 • Exam diagram & MCQ practice

Key exam idea: Electrolytes conduct through mobile ions; electrolysis uses electrical energy to drive a non-spontaneous chemical change.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which particles carry charge through an electrolyte solution?

A. mobile ions
B. neutral atoms only
C. photons
D. neutrons

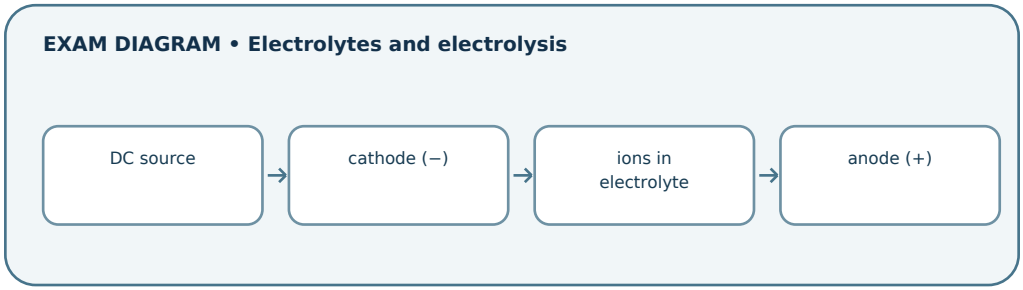
Answer: A. Cations and anions move through the solution and carry electric current.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.07 Electrolytic cell components

Core concept

An electrolytic cell contains an electrolyte, two electrodes and an external direct-current source. The power supply drives electron flow through the external circuit and ion movement within the electrolyte. In an electrolytic cell the cathode is negative and the anode is positive. Do not confuse this polarity with a galvanic cell.

Remember

Electrolytic: anode (+), cathode (–)

Electrolytic cell components	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Draw and label a simple electrolytic cell.

B. Explain why the power source is required for electrolysis.

2.07 • Exam diagram & MCQ practice

Key exam idea: An electrolytic cell has an electrolyte, two electrodes and an external DC power source.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In an electrolytic cell, the power supply is used to:

A. drive a non-spontaneous reaction
B. measure pH
C. remove all ions from solution
D. create a salt bridge

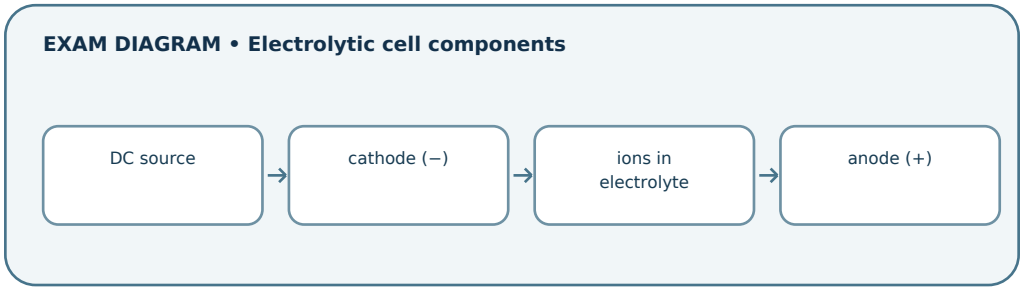
Answer: A. The external source supplies electrical energy to force the redox reaction to occur.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.08 Preferential discharge

Core concept

When several ions are present, the ion discharged at an electrode depends on factors such as position in the electrochemical series, ion concentration and electrode nature, as treated in the textbook. Aqueous solutions also contain water-derived H■/OH■ species, so the products may not simply be the salt ions. State the solution and electrode conditions before predicting products.

Remember

Competing ions → selective discharge

Preferential discharge	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. What factors influence preferential discharge during electrolysis?

B. Why may an aqueous salt solution produce a gas instead of depositing its metal?

2.08 • Exam diagram & MCQ practice

Key exam idea: Discharge depends on ion identity, concentration and electrode conditions; in aqueous solutions water may compete with metal ions.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why may hydrogen form instead of a reactive metal during aqueous electrolysis?

A. Water/H^+ may be preferentially reduced
B. The cathode only attracts anions
C. Electrons cannot enter the solution
D. All metal ions are neutral

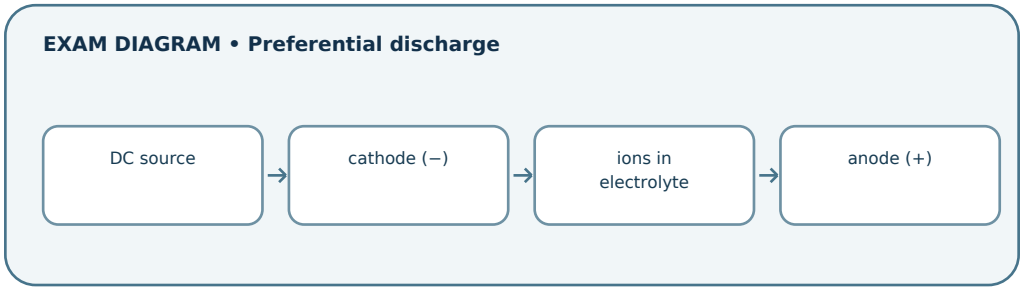
Answer: A. At the cathode, reduction competition can favour water or hydrogen ions over deposition of a very reactive metal.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.09 Electrolysis of aqueous solutions

Core concept

Predict cathode and anode products separately. At the cathode, reduction occurs; at the anode, oxidation occurs. Consider the dissolved ions, water, concentration and electrode material. Then write the half-equations and combine them to obtain the overall reaction. The textbook examines selected aqueous electrolytes as worked cases.

Remember

Cathode reduction / anode oxidation

Electrolysis of aqueous solution	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Outline how to predict products when an aqueous electrolyte is electrolyzed.

B. Write the half-reactions for electrolysis of an aqueous solution when hydrogen forms at the cathode and oxygen at the anode.

2.09 • Exam diagram & MCQ practice

Key exam idea: Reduction occurs at the cathode and oxidation at the anode; products must be predicted separately.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. At which electrode does reduction occur during electrolysis?

A. cathode
B. anode
C. salt bridge
D. power switch

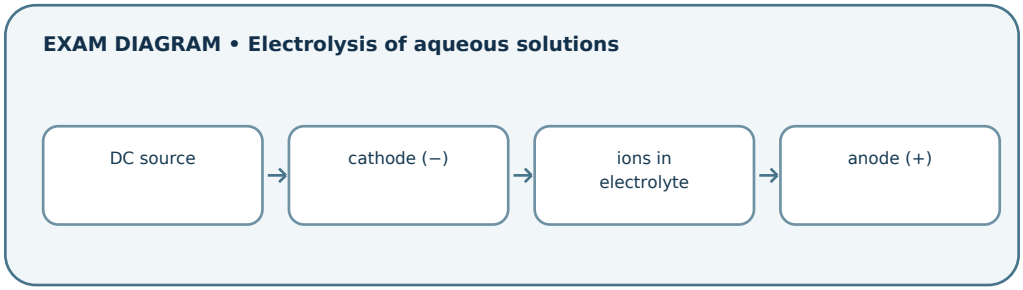
Answer: A. Reduction is gain of electrons, and the cathode is the electrode where reduction takes place.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.10 Electrode reactions and half-equations

Core concept

A half-equation shows the electron transfer at one electrode. Electrons appear on the left for reduction and on the right for oxidation. Atoms and charge must balance. Multiplying half-equations may be necessary before adding them so that electrons cancel. Always check the final equation for both mass and charge conservation.

Remember

Reduction: Ox + ne■ → Red

Electrode reactions and half-equ	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the cathode half-equation for Cu^{2}■ + 2e■ → Cu and classify it.

B. Write an oxidation half-equation for a metal M forming M^{2}■.

2.10 • Exam diagram & MCQ practice

Key exam idea: A reduction half-equation has electrons on the left; an oxidation half-equation has electrons on the right.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which half-equation represents reduction?

A. $Cu^{2+} + 2e^- \rightarrow Cu$
B. $Zn \rightarrow Zn^{2+} + 2e^-$
C. $2Cl^- \rightarrow Cl_2 + 2e^-$
D. $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

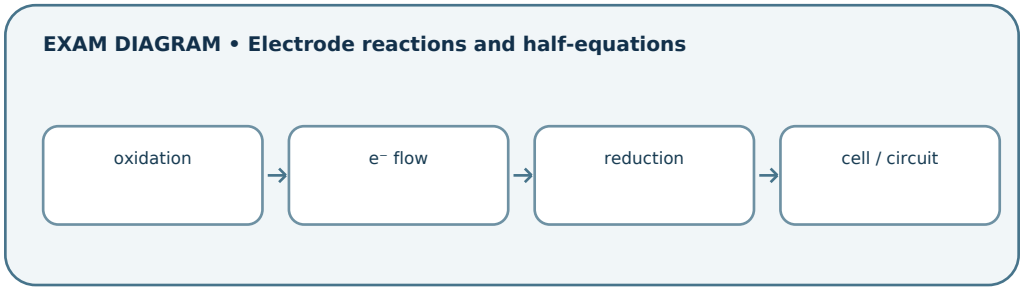
Answer: A. Copper ions gain two electrons to form copper metal.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.11 Faraday’s first law

Core concept

Faraday’s first law states that the mass of substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte. Charge depends on current and time. Thus, for the same electrolyte and conditions, increasing current or time increases the deposited mass.

Remember

$Q = It$; $m \propto Q$

Faraday’s first law	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. State Faraday’s first law in words.

B. What happens to deposited mass if current is doubled while time is unchanged?

2.11 • Exam diagram & MCQ practice

Key exam idea: The mass deposited or liberated is directly proportional to the quantity of electricity passed.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. If current is doubled while time remains constant, the deposited mass ideally:

A. doubles
B. halves
C. remains unchanged
D. becomes zero

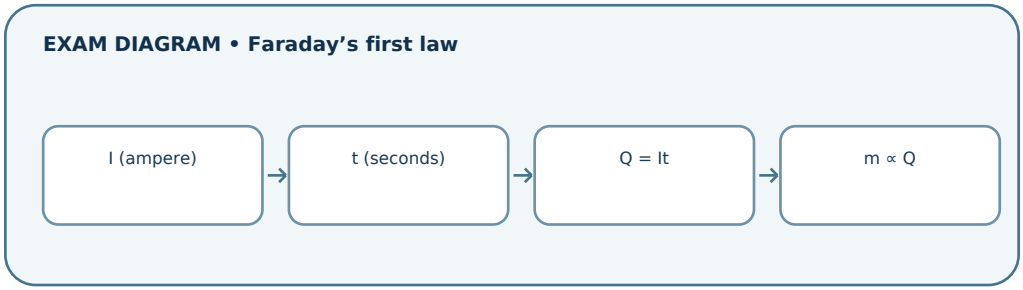
Answer: A. Q = It, so doubling current doubles charge; Faraday’s first law makes deposited mass double.

Question 2. In the diagram, which labelled step comes immediately after “I (ampere)”?

A. t (seconds)
B. Q = It
C. m ∝ Q
D. I (ampere)

Answer: A. The schematic is arranged in sequence: I (ampere) → t (seconds) → Q = It → m ∝ Q.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.12 Faraday constant and charge

Core concept

One mole of electrons carries approximately 96,500 coulombs. The charge passed is current multiplied by time, with time expressed in seconds. Convert charge to moles of electrons, then use the electrode half-equation to relate electron amount to product amount.

Remember

Q = It; n(e■) = Q/F; F ≈ 96,500 C mol■¹

Faraday constant and charge	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Calculate charge passed by a 2.0 A current for 10 minutes.

B. How many moles of electrons correspond to 19,300 C?

2.12 • Exam diagram & MCQ practice

Key exam idea: Charge passed is Q = It; one mole of electrons carries approximately 96,500 C.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. What charge passes when a 2.0 A current flows for 10 minutes?

A. 1200 C
B. 20 C
C. 600 C
D. 96500 C

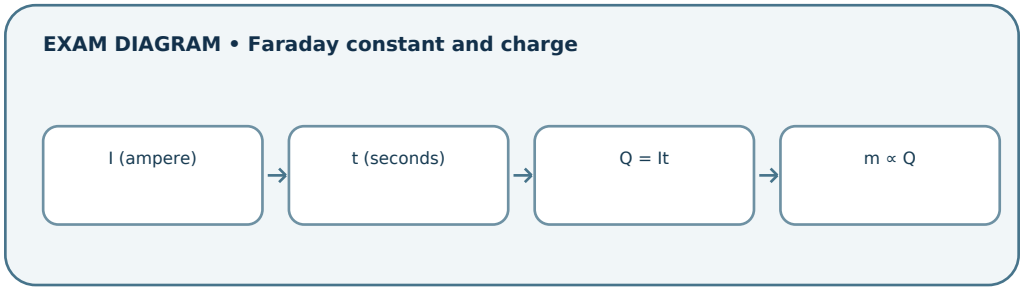
Answer: A. Convert 10 minutes to 600 s. Q = It = 2.0 × 600 = 1200 C.

Question 2. In the diagram, which labelled step comes immediately after “I (ampere)”?

A. t (seconds)
B. Q = It
C. m ∝ Q
D. I (ampere)

Answer: A. The schematic is arranged in sequence: I (ampere) → t (seconds) → Q = It → m ∝ Q.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.13 Electrolysis mass calculations

Core concept

Use a sequence: calculate Q = It; calculate moles of electrons Q/F; use the balanced half-equation to find moles of product; convert to mass using molar mass. Include the electron-to-product mole ratio. Check units and significant figures, and use the correct electrode reaction.

Remember

m = (MIt)/(nF)

Electrolysis mass calculations	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Calculate the mass of a metal deposited by a known current and time when its ion needs two electrons.

B. Explain why the electron number n appears in the mass equation.

2.13 • Exam diagram & MCQ practice

Key exam idea: Use Q = It, moles of electrons = Q/F, then apply the balanced electrode half-equation.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. For $M^{2+} + 2e^- \rightarrow M$, how many moles of electrons are needed to deposit 1 mol M?

A. 2 mol
B. 1 mol
C. 3 mol
D. 0.5 mol

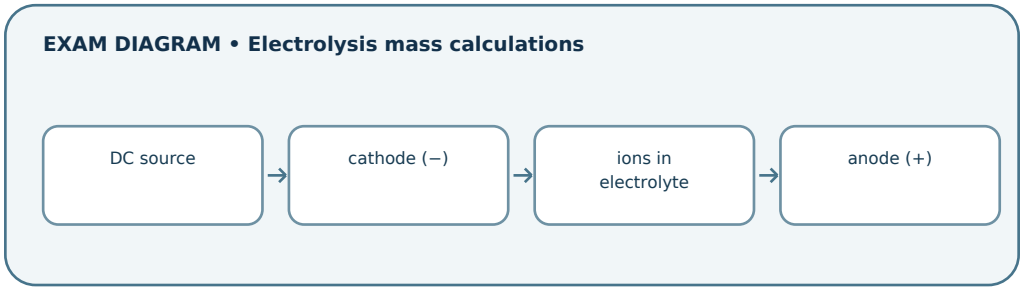
Answer: A. The half-equation shows that each M^{2+} ion requires two electrons.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.14 Faraday’s second law

Core concept

Faraday’s second law compares substances deposited by the same quantity of electricity. Their masses are proportional to their equivalent masses. Equivalent mass depends on molar mass and the number of electrons involved per formula unit. Use the balanced electrode reaction to determine the electron number.

Remember

m■/m■ = E■/E■

Faraday’s second law	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. State Faraday’s second law.

B. Compare the masses of two metals deposited by equal charge using their equivalent masses.

2.14 • Exam diagram & MCQ practice

Key exam idea: For the same charge, deposited masses are proportional to their chemical equivalent masses.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. When equal charge passes through two electrolytes, deposited masses are proportional to:

A. molar mass divided by electron number
B. solution volume
C. electrode area only
D. atomic number alone

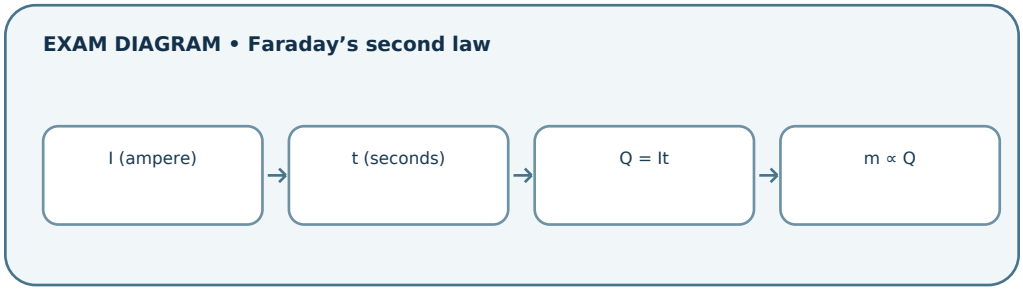
Answer: A. Equivalent mass is molar mass divided by the number of electrons exchanged per ion.

Question 2. In the diagram, which labelled step comes immediately after “I (ampere)”?

A. t (seconds)
B. Q = It
C. m ∝ Q
D. I (ampere)

Answer: A. The schematic is arranged in sequence: I (ampere) → t (seconds) → Q = It → m ∝ Q.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.15 Industrial electrolysis

Core concept

Electrolysis is applied in metal extraction, purification and manufacture of useful chemicals. Industrial processes depend on electrode reactions, electrolyte composition, energy use and product separation. In explaining an application, identify the feed material, the electrode products and the purpose of the process. Include energy and environmental considerations where relevant.

Remember

Industrial use: extraction • refining • chemical production

Industrial electrolysis	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Describe one industrial application of electrolysis from the textbook.

B. Why is electrolysis useful for purifying a metal?

2.15 • Exam diagram & MCQ practice

Key exam idea: Electrolysis is used in metal extraction, purification and manufacture of chemicals such as chlorine and sodium hydroxide.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is an industrial application of electrolysis?

A. extraction of aluminium
B. fractional distillation of crude oil
C. filtration of sand
D. crystallization of sugar

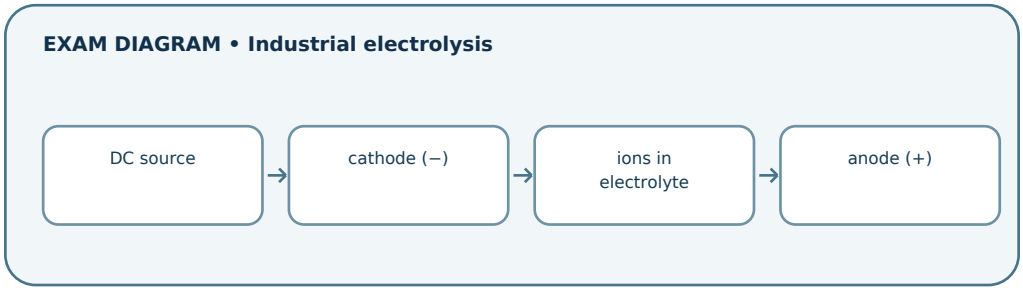
Answer: A. Aluminium is extracted from molten alumina by an electrolytic process.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.16 Electrolytic refining

Core concept

In electrolytic refining, impure metal is commonly used as the anode and a thin sheet of pure metal as the cathode; a suitable electrolyte contains ions of the metal. Metal dissolves from the anode and deposits on the cathode. Insoluble impurities may collect as anode sludge. Explain the movement of metal ions and electron transfer.

Remember

Anode: M → M■■ + ne■; cathode: M■■ + ne■ → M

Electrolytic refining	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain how copper can be purified electrolytically.

B. Identify the roles of the impure anode, pure cathode and electrolyte.

2.16 • Exam diagram & MCQ practice

Key exam idea: In copper refining, impure copper is the anode, pure copper is the cathode, and copper salt solution is the electrolyte.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. During electrolytic refining of copper, the impure copper is connected as the:

A. anode
B. cathode
C. salt bridge
D. insulator

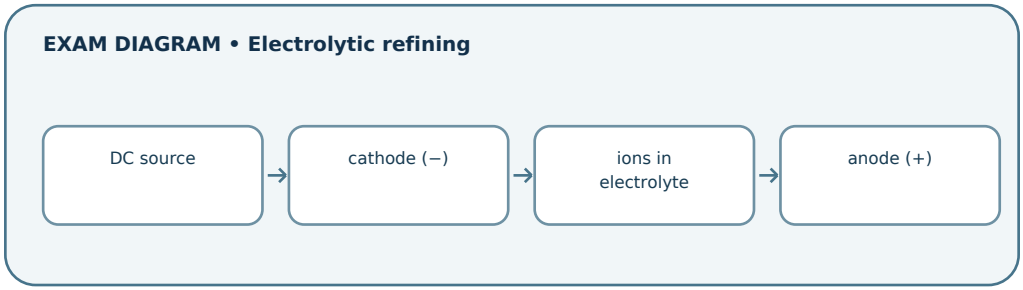
Answer: A. Copper atoms from the impure anode oxidize to Cu^{2+}; copper deposits on the pure cathode.

Question 2. In the diagram, which labelled step comes immediately after “DC source”?

A. cathode (−)
B. ions in electrolyte
C. anode (+)
D. DC source

Answer: A. The schematic is arranged in sequence: DC source → cathode (−) → ions in electrolyte → anode (+).

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.17 Voltaic (galvanic) cells

Core concept

A voltaic cell converts chemical energy from a spontaneous redox reaction into electrical energy. Oxidation occurs at the anode and reduction at the cathode. Electrons flow through the external circuit from anode to cathode; ions move through the electrolyte or salt bridge to maintain charge balance. The cell reaction is spontaneous.

Remember

Anode = oxidation; cathode = reduction; e■ flow anode → cathode

Voltaic (galvanic) cells	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the energy conversion in a voltaic cell.

B. Draw a labelled cell and show electron flow.

2.17 • Exam diagram & MCQ practice

Key exam idea: A voltaic cell converts spontaneous redox chemical energy into electrical energy; oxidation occurs at the anode.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A galvanic cell converts:

A. chemical energy into electrical energy
B. electrical energy into chemical energy
C. heat into nuclear energy
D. light into sound

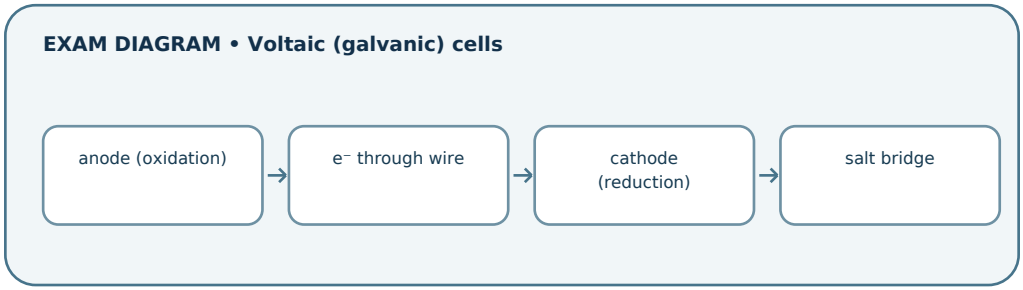
Answer: A. A spontaneous redox reaction drives electron flow through an external circuit.

Question 2. In the diagram, which labelled step comes immediately after “anode (oxidation)”?

A. e^- through wire
B. cathode (reduction)
C. salt bridge
D. anode (oxidation)

Answer: A. The schematic is arranged in sequence: anode (oxidation) → e^- through wire → cathode (reduction) → salt bridge.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.18 Cell notation and salt bridge

Core concept

Cell notation summarizes the two half-cells. A vertical line represents a phase boundary and a double line represents the salt bridge. The salt bridge completes the internal ionic circuit and prevents charge build-up. It does not carry electrons; electrons travel through the external wire.

Remember

Anode | anode solution || cathode solution | cathode

Cell notation and salt bridge	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the function of the salt bridge.

B. Interpret a cell diagram written as Zn|Zn²■||Cu²■|Cu.

2.18 • Exam diagram & MCQ practice

Key exam idea: In cell notation, a single line marks a phase boundary and a double line marks the salt bridge; the bridge maintains charge balance.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In $Zn|Zn^{2+}||Cu^{2+}|Cu$, the double vertical line represents the:

A. salt bridge
B. metal electrode
C. electron path
D. gas outlet

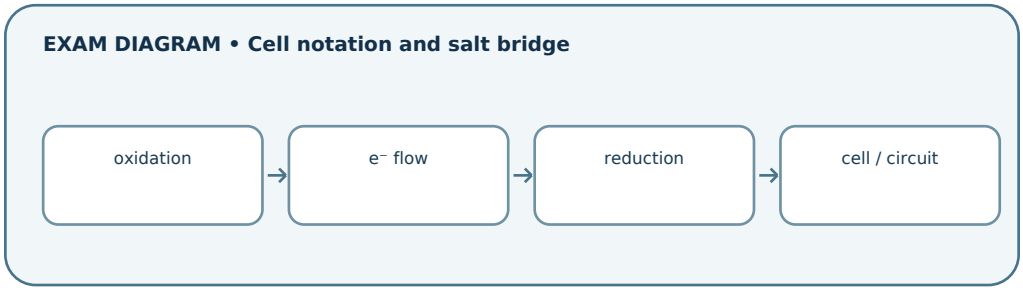
Answer: A. The double line conventionally represents the salt bridge or porous separator between half-cells.

Question 2. In the diagram, which labelled step comes immediately after “oxidation”?

A. e^- flow
B. reduction
C. cell / circuit
D. oxidation

Answer: A. The schematic is arranged in sequence: oxidation → e^- flow → reduction → cell / circuit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.19 Cell potential

Core concept

The cell potential expresses the driving force for electron flow. Standard cell potential is obtained by combining standard reduction potentials, using cathode minus anode reduction potential. A positive standard cell potential indicates a spontaneous cell reaction under standard conditions. Use the reduction-potential table consistently.

Remember

$E°_{cell} = E°_{cathode} - E°_{anode}$

Cell potential	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Calculate $E°_{cell}$ from two standard reduction potentials.

B. What does a positive $E°_{cell}$ indicate about a cell reaction under standard conditions?

2.19 • Exam diagram & MCQ practice

Key exam idea: E°cell = E°cathode − E°anode using standard reduction potentials; a positive value indicates a spontaneous cell reaction under standard conditions.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A positive standard cell potential indicates that the cell reaction is:

A. spontaneous under standard conditions
B. non-spontaneous under all conditions
C. at equilibrium only
D. impossible

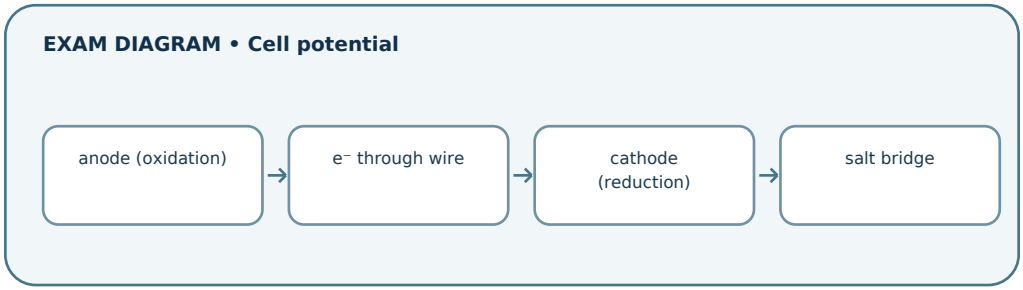
Answer: A. A positive E°cell corresponds to a negative standard Gibbs energy change for the cell reaction.

Question 2. In the diagram, which labelled step comes immediately after “anode (oxidation)”?

A. e^- through wire
B. cathode (reduction)
C. salt bridge
D. anode (oxidation)

Answer: A. The schematic is arranged in sequence: anode (oxidation) → e^- through wire → cathode (reduction) → salt bridge.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



2.20 Voltaic cells, corrosion and impact

Core concept

The textbook connects galvanic cells with common cells and corrosion. Corrosion is an electrochemical process in which a metal is oxidized by its environment. Explain the anodic and cathodic regions and how contact with another metal or environmental conditions can influence corrosion. Discuss useful cells alongside their environmental and social effects.

Remember

Corrosion involves metal oxidation and coupled reduction

Voltaic cells, corrosion and imp	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the electrochemical basis of iron corrosion.

B. Compare a useful voltaic cell with an unwanted corrosion process.

2.20 • Exam diagram & MCQ practice

Key exam idea: Corrosion is an unwanted electrochemical oxidation process; iron rusting involves anodic iron oxidation and cathodic oxygen reduction.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process is the main anodic change during iron corrosion?

A. $Fe \rightarrow Fe^{2+} + 2e^-$
B. $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
C. $Fe^{2+} + 2e^- \rightarrow Fe$
D. $2H^+ + 2e^- \rightarrow H_2$

Answer: A. Iron atoms lose electrons to form Fe^{2+} at anodic regions; this is oxidation.

Question 2. In the diagram, which labelled step comes immediately after “anode (oxidation)”?

A. e^- through wire
B. cathode (reduction)
C. salt bridge
D. anode (oxidation)

Answer: A. The schematic is arranged in sequence: anode (oxidation) → e^- through wire → cathode (reduction) → salt bridge.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.

